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Experiment-3 : Data Ingestion and Cleaning

Aim:- Data Wrangling with Pandas: Manipulate and transform data to suit the analysis need.

Q1. What is data wrangling?. Explain the difference between Data Wrangling and Data preprocessing.

ANS:- Data wrangling is the process of cleaning, transforming, and organizing raw data into a more useful and structured format for analysis. It involves tasks such as handling missing values, filtering data, and reshaping datasets. Data preprocessing, on the other hand, is a broader term that includes all steps taken before applying machine learning algorithms, including data wrangling, feature scaling, encoding categorical data, and splitting datasets. In simple terms, data wrangling mainly focuses on cleaning and transforming data, while data preprocessing includes additional steps required to prepare data specifically for modeling.

```
import pandas as pd
data = pd.DataFrame({
    'Name': ['Ali', 'Sara', None, 'John'],
    'Age': [20, None, 22, 21],
    'Marks': [85, 90, None, 88]
})
print("Original Data:")
print(data)
data['Age'].fillna(data['Age'].mean(), inplace=True)
data['Marks'].fillna(0, inplace=True)
data.dropna(subset=['Name'], inplace=True)
print("\nAfter Data Wrangling:")
print(data)
from sklearn.preprocessing import MinMaxScaler
scaler = MinMaxScaler()
data[['Marks']] = scaler.fit_transform(data[['Marks']])
print("\nAfter Data Preprocessing (Scaling):")
print(data)
```

Original Data:

	Name	Age	Marks
0	Ali	20.0	85.0
1	Sara	NaN	90.0
2	None	22.0	NaN
3	John	21.0	88.0

After Data Wrangling:

	Name	Age	Marks
0	Ali	20.0	85.0
1	Sara	21.0	90.0
3	John	21.0	88.0

/tmp/ipykernel_27644/566247699.py:9: FutureWarning: A value is trying to be set on a copy of a Dat
The behavior will change in pandas 3.0. This inplace method will never work because the intermedia

For example, when doing 'df[col].method(value, inplace=True)', try using 'df.method({col: value},

```
data['Age'].fillna(data['Age'].mean(), inplace=True)
```

/tmp/ipykernel_27644/566247699.py:10: FutureWarning: A value is trying to be set on a copy of a Da

The behavior will change in pandas 3.0. This inplace method will never work because the intermedia

For example, when doing 'df[col].method(value, inplace=True)', try using 'df.method({col: value},

```
data['Marks'].fillna(0, inplace=True)
```

After Data Preprocessing (Scaling):

	Name	Age	Marks
0	Ali	20.0	0.0
1	Sara	21.0	1.0
3	John	21.0	0.6

Q2. What is the importance of exploratory data analysis(EDA) in pandas?

ANS:- Exploratory Data Analysis (EDA) is the important step in Data analysis that help in understanding the structure, pattern, and relationship with dataset. In Pandas, EDA allow user to summarize the data using the statistical measure, visilize trends, and detect anomalies and outliers. It help in identifying missing value, understanding distrubution, and making inform decisions about further data processing. EDA is essentisl beacuse it provide insights into the data before applying any complex analysis or machine learning models

```
import pandas as pd
data = pd.DataFrame({
    'Name': ['Ali', 'Sara', 'John', 'Aman'],
    'Age': [20, 21, 22, 23],
    'Marks': [85, 90, 88, 92]
})
print(data.info())
print(data.describe())
```

```
<class 'pandas.core.frame.DataFrame'>
RangeIndex: 4 entries, 0 to 3
Data columns (total 3 columns):
#   Column  Non-Null Count  Dtype
---  -
0   Name    4 non-null         object
1   Age     4 non-null         int64
2   Marks   4 non-null         int64
dtypes: int64(2), object(1)
memory usage: 228.0+ bytes
None
```

	Age	Marks
count	4.000000	4.000000
mean	21.500000	88.750000
std	1.290994	2.986079
min	20.000000	85.000000
25%	20.750000	87.250000
50%	21.500000	89.000000
75%	22.250000	90.500000
max	23.000000	92.000000

Q3.What are the main differences between series, Dataframe,and Panel?

```
import pandas as pd

# Create a Series
s_data = [10, 20, 30, 40, 50]
s_index = ['a', 'b', 'c', 'd', 'e']
series_example = pd.Series(s_data, index=s_index)

print("Pandas Series Example:")
print(series_example)
print(f"\nType: {type(series_example)}")
print(f"Dimensions: {series_example.ndim}")
```

Pandas Series Example:

```
a    10
b    20
c    30
d    40
e    50
dtype: int64
```

Type: <class 'pandas.core.series.Series'>
Dimensions: 1

```
# Create a DataFrame
df_data = {
    'Name': ['Alice', 'Bob', 'Charlie', 'David'],
    'Age': [25, 30, 35, 40],
    'City': ['New York', 'London', 'Paris', 'Tokyo']
}
dataframe_example = pd.DataFrame(df_data)

print("\nPandas DataFrame Example:")
print(dataframe_example)
print(f"\nType: {type(dataframe_example)}")
print(f"Dimensions: {dataframe_example.ndim}")
```

Pandas DataFrame Example:

	Name	Age	City
0	Alice	25	New York
1	Bob	30	London
2	Charlie	35	Paris
3	David	40	Tokyo

Type: <class 'pandas.core.frame.DataFrame'>
Dimensions: 2

4.Explain the use of:

a)isnull(): The isnull() function is used to check the missing or null value in a dataset. It returns a boolean result where the true indicates a missing value and indicates a non-missing values

```
import pandas as pd
df = pd.DataFrame({'A' : [1, None, 3]})
print(df.isnull())
```

	A
0	False
1	True
2	False

c)dropna():- the dropna() function is used to remove rows or columns that contain missing values. By default, it remove rows with at least one null value.

```
import pandas as pd
df = pd.DataFrame({'A' : [1, None, 3]})
print(df.dropna())
```

	A
0	1.0
2	3.0

d)fillna():- The fillna() function is used to replace the missing values with a specified values such as zero, mean, or any constant.

```
import pandas as pd
df = pd.DataFrame({'A': [1, None, 3]})
print(df.fillna(0))
```

```
   A
0  1.0
1  0.0
2  3.0
```

Q5. what is column transformation and why is it important in data analysis?

ANS:- Column transformation refers to modifying or converting the values of the column to make them suitable for analysis. This may include data types, scaling values, applying mathematical operations, or creating new derived column. Column transformation is important because it improves the quality and usability of the data, making it easier to analyse and interpret. It also plays a key role in preparing data for machine learning models, where properly formatted data leads to better performance and accurate results.

```
import pandas as pd
df = pd.DataFrame({'A': [1, None, 3]})
print(df.isnull())
print(df.notnull())
print(df.dropna())
print(df.fillna(0))
```

```
   A
0  False
1   True
2  False
   A
0   True
1  False
2   True
   A
0  1.0
2  3.0
   A
0  1.0
1  0.0
2  3.0
```

Conclusion:-

Data wrangling using Pandas is a crucial step in data analysis that helps convert raw data into a clean and structured format. Techniques like EDA, handling missing values, and column transformation improve data quality and make the analysis more accurate and reliable.

